Autonomous line-following Material transport robot

1. Process Sheet

1. REQUIREMENT ENGINEERING

Process Step	Input	Tools / Methods	Critical Parameters	Output
Plan – Identify need	Industry need, logistics	Observation	Load type, path	Problem statement
Plan – Gather requirements	User needs	Interview, survey	Speed, cost	Requirement list
Do – Define specifications	Requirements	SRS	Accuracy, payload	System specs
Check – Validate requirements	SRS	Review	Completeness	Approved SRS
Act – Refine requirements	Feedback	Gap analysis	Feasibility	Updated SRS
Act – Freeze requirements	Final SRS	Approval	Stability	Requirement freeze

2. CONCEPT DESIGN

Process Step	Input	Tools / Methods	Critical Parameters	Output
Plan – Generate concepts	Requirements	Brainstorming	Line sensing type	Concept ideas
Plan – Compare concepts	Concepts	Pugh matrix	Cost vs performance	Best concept
Do – Develop block diagram	Selected idea	Functional diagram	Control flow	Block diagram
Check – Evaluate concept	Model	Simulation	Accuracy	Evaluated concept

Act – Improve concept	Feedback	Refinement	Simplicity	Improved concept
Act – Freeze concept	Final concept	Approval	Feasibility	Concept freeze

3. EMBODIMENT DESIGN

Process Step	**Input**	**Tools / Methods**	**Critical Parameters**	**Output**
Plan – Define architecture	Concept	System design	Subsystem integration	Layout
Plan – Select components	Sensors, motors	Datasheets	Compatibility	Component list
Do – Design layout	Components	CAD	Size, alignment	Layout model
Check – Verify design	Layout	Fit check	Stability	Verified layout
Act – Optimize design	Review	Optimization	Weight, efficiency	Optimized layout
Act – Freeze embodiment	Final layout	Approval	Reliability	Design freeze

4. DETAIL DESIGN

Process Step	**Input**	**Tools / Methods**	**Critical Parameters**	**Output**
Plan – Define detail scope	Layout	Planning	Dimensions	Detail plan
Plan – Specify parts	Components	Datasheets	Tolerance	Spec sheet
Do – Draw circuits & CAD	Specs	CAD, PCB design	Accuracy	Detailed drawings

Check – Review drawings	Drawings	Checklist	Errors	Review report
Act – Correct design	Feedback	Editing	Compliance	Updated design
Act – Release design	Final design	Approval	Manufacturability	Design release

5. SOURCING

Process Step	Input	Tools / Methods	Critical Parameters	Output
Plan – Prepare BOM	Design	BOM sheet	Cost	BOM
Plan – Select suppliers	Market data	RFQ	Quality	Supplier list
Do – Purchase parts	Suppliers	Procurement	Specs match	Components
Check – Inspect parts	Received parts	Testing	Quality	Inspection report
Act – Replace defects	Inspection	Correction	Compliance	Accepted parts
Act – Approve sourcing	Final parts	Approval	Availability	Sourcing complete

6. PROTOTYPE BUILDING

Process Step	Input	Tools / Methods	Critical Parameters	Output
Plan – Build plan	Design	Planning	Assembly order	Plan
Plan – Setup tools	Materials	Checklist	Safety	Ready setup
Do – Assemble robot	Components	Wiring, assembly	Fit	Prototype V1

Check – Inspect build	Prototype	Testing	Stability	Inspection
Act – Fix issues	Errors	Rework	Reliability	Prototype V2
Act – Finalize build	Improved model	Approval	Functionality	Final prototype

7. PROTOTYPE TESTING

Process Step	Input	Tools / Methods	Critical Parameters	Output
Plan – Test plan	Prototype	Planning	Path type	Test plan
Plan – Setup test	Track, sensors	Calibration	Accuracy	Test setup
Do – Run robot	Prototype	Experiment	Speed, tracking	Test data
Check – Analyze results	Data	Analysis	Error rate	Report
Act – Identify issues	Report	Root cause	Faults	Improvement list
Act – Approve test	Final data	Review	Acceptance	Test approval

8. ITERATION FOR OPTIMIZATION

Process Step	Input	Tools / Methods	Critical Parameters	Output
Plan – Identify issues	Test report	Analysis	Error margin	Plan
Plan – Select improvements	Options	Decision matrix	Cost	Strategy
Do – Implement changes	Design	Modification	Performance	Improved system

Check – Retest	System	Testing	Efficiency	Report
Act – Fine tuning	Feedback	Adjustment	Stability	Optimized robot
Act – Freeze design	Final system	Approval	Performance	Final design

9. DOCUMENTING FOR SHARING

Process Step	Input	Tools / Methods	Critical Parameters	Output
Plan – Identify documents	Data	Planning	Completeness	Doc list
Plan – Organize data	Files	Structuring	Clarity	Draft
Do – Prepare report	Data	Word, diagrams	Accuracy	Report
Check – Review report	Draft	Proofreading	Errors	Feedback
Act – Correct report	Feedback	Editing	Quality	Final report
Act – Publish project	Final doc	Submission	Accessibility	Project submission

2. ACTIVITY DURATION TABLE (200 HOURS)

S.No	Activity Description	Duration (Hours)
1	Identify problem & objective	3
2	Collect requirements	4
3	Identify system parameters	3
4	Define system requirements	4
5	Review requirements	3
6	Freeze requirements	3
7	Generate concept ideas	4
8	Compare concepts	4
9	Develop block diagram	4
10	Evaluate concepts	4
11	Improve concept	3
12	Freeze concept	3
13	Define system architecture	4
14	Select sensors & motors	5
15	Design layout	5
16	Verify layout	4
17	Optimize layout	4
18	Freeze embodiment design	4
19	Define detail scope	4
20	Prepare specifications	5

21	Create circuit & CAD drawings	7
22	Review drawings	4
23	Correct design	4
24	Release design	3
25	Prepare BOM	3
26	Select suppliers	4
27	Purchase components	5
28	Inspect components	4
29	Resolve defects	3
30	Approve sourcing	3
31	Plan prototype build	3
32	Prepare tools & setup	3
33	Assemble robot	7
34	Inspect prototype	4
35	Rework issues	5
36	Finalize prototype	3
37	Prepare testing plan	3
38	Setup testing environment	3
39	Conduct testing	6
40	Analyze test data	4
41	Identify faults	3
42	Approve test results	2

43	Identify optimization areas	3
44	Select optimization method	3
45	Implement improvements	5
46	Retest system	3
47	Fine tuning	3
48	Freeze optimized design	3
49	Identify documentation needs	3
50	Organize data	3
51	Prepare report	3
52	Review documentation	2
53	Final corrections	4
54	Submit project	2
TOTAL		**200**

3. ACTIVITY PRECEDENCE TABLE

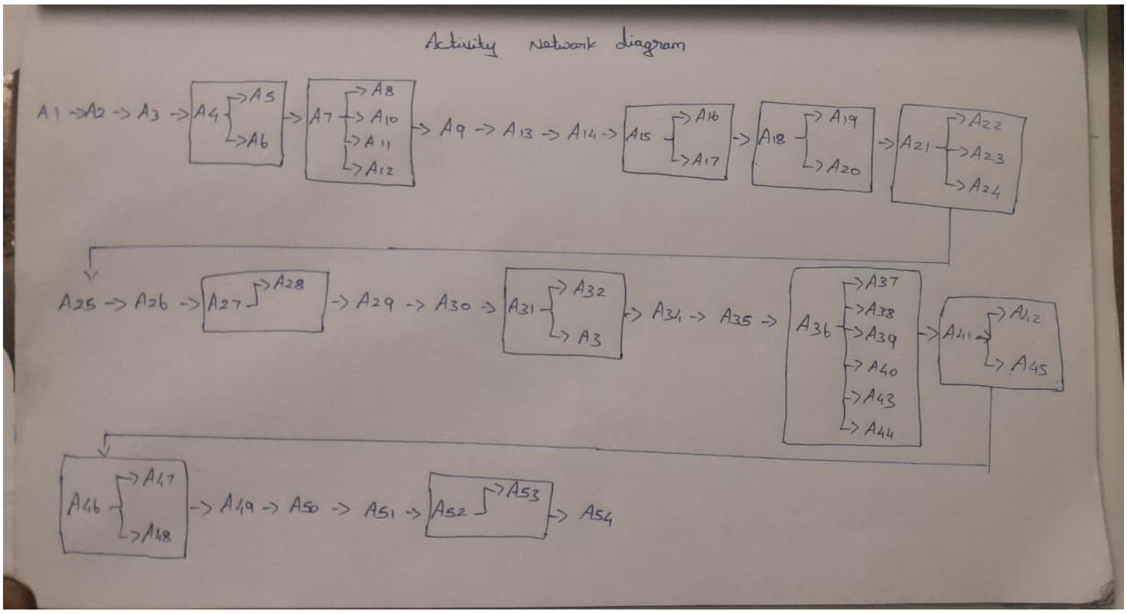
Activity No.	Activity Description	Immediate Predecessor
1	Identify problem & objective	—
2	Collect requirements	1
3	Identify system parameters	2
4	Define system requirements	3

5	Review requirements	4
6	Freeze requirements	5
7	Generate concept ideas	6
8	Compare concepts	7
9	Develop block diagram	8
10	Evaluate concepts	9
11	Improve concept	10
12	Freeze concept	11
13	Define system architecture	12
14	Select sensors & motors	13
15	Design layout	14
16	Verify layout	15
17	Optimize layout	16
18	Freeze embodiment design	17
19	Define detail scope	18
20	Prepare specifications	19
21	Create circuit & CAD drawings	20
22	Review drawings	21
23	Correct design	22
24	Release design	23
25	Prepare BOM	24
26	Select suppliers	25
27	Purchase components	26
28	Inspect components	27

29	Resolve defects	28
30	Approve sourcing	29
31	Plan prototype build	30
32	Prepare tools & setup	31
33	Assemble robot	32
34	Inspect prototype	33
35	Rework issues	34
36	Finalize prototype	35
37	Prepare testing plan	36
38	Setup testing environment	37
39	Conduct testing	38
40	Analyze test data	39
41	Identify faults	40
42	Approve test results	41
43	Identify optimization areas	42
44	Select optimization method	43
45	Implement improvements	44
46	Retest system	45
47	Fine tuning	46
48	Freeze optimized design	47
49	Identify documentation needs	48
50	Organize data	49
51	Prepare report	50
52	Review documentation	51

53	Final corrections	52
54	Submit project	53

Activity Network Diagram



Series and Parallel Path